Logarithmic algebraic geometry: motivation

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1 Euclidean topology

Let $\mathbb{C}$ be the field of complex numbers, and let $\mathbb{A}^n = \operatorname{Spec}\mathbb{C}[x_1, \ldots, x_n]$. Recall that $\mathbb{A}^n$ contains not only the points $\mathbb{C}^n = \{(a_1, \ldots, a_n) \mid a_i \in \mathbb{C}\}$ but also points corresponding to positive-dimensional irreducible closed subschemes of $\mathbb{A}^n$, and that $\mathbb{A}^n$ carries a rather coarse topology, the *Zariski topology*. For a closed subscheme

$$X = \{f_1 = \ldots = f_r = 0\} \subset \mathbb{A}^n,$$

we write

$$X^{\mathrm{top}} = \{(a_1, \ldots, a_n) \mid \forall\, j : f_j(a_1, \ldots, a_n) = 0\} \subset \mathbb{C}^n$$

for the corresponding subset of $\mathbb{C}^n$, endowed with the topology induced by the *Euclidean topology* on $\mathbb{C}^n$. This construction extends to a functor

$$(-)^{\mathrm{top}} \colon \ \mathrm{Sch}^{\mathrm{lft}}_{\mathbb{C}} \to \mathrm{Top},$$

where $\mathrm{Sch}^{\mathrm{lft}}_{\mathbb{C}}$ is the category of schemes locally of finite type over $\operatorname{Spec}\mathbb{C}$. We say that X^{top} is the *underlying Euclidean topological space* of X.

Example 1.1. Let $X = \mathbb{A}^1$. Then, we have $X = \mathbb{C} \sqcup \{\eta\}$ set-theoretically, and the typical closed subset is a finite subset of $\mathbb{C}$. On the other hand, we have $X^{\mathrm{top}} = \mathbb{C}$ set-theoretically, and the typical closed subset is a closed disk $\{z \in \mathbb{C} \mid |z - z_0| \leq \delta\}$. ♢

2 Calabi–Yau manifolds

Let $X \to \operatorname{Spec}\mathbb{C}$ be smooth, proper, and irreducible, and let $d = \dim(X)$. Then, the sheaf of Kähler differential forms Ω^1_X is a vector bundle of rank d. We have the *canonical sheaf*

$$\Omega^d_X \coloneqq \bigwedge_{\mathcal{O}_X}^{d} \Omega^1_X,$$

which is a line bundle. It is an important invariant of X.

Definition 2.1. An *algebraic Calabi–Yau manifold* is a smooth and proper irreducible scheme $X \to \operatorname{Spec}\mathbb{C}$ such that $\Omega^d_X \cong \mathcal{O}_X$ for $d = \dim(X)$. An algebraic Calabi–Yau manifold is *strict* if $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < d$. ⧫

Remark 2.2. Let X be an algebraic Calabi–Yau manifold. Since X is irreducible, we have $H^0(X, \mathcal{O}_X) \cong \mathbb{C}^1$. Serre duality gives

$$H^d(X, \mathcal{O}_X) \cong H^0(X, \Omega_X^d)^\vee,$$

so the Calabi–Yau condition yields $H^d(X, \mathcal{O}_X) \cong \mathbb{C}^1$. ◊

To construct our first examples of algebraic Calabi–Yau manifolds, we use:

Fact 2.3. *Let $X = \{f = 0\} \subset \mathbb{P}^{d+1}$ be a smooth hypersurface cut out by a homogeneous polynomial $f \in \mathbb{C}[x_0, \dots, x_{d+1}]$. Then, we have $\Omega_X^d \cong \mathcal{O}_X(-d-2+\delta)$ for $\delta = \deg(f)$.*

Lemma 2.4. *Let $d \geq 2$, and let $X = \{f = 0\} \subset \mathbb{P}^{d+1}$ be a smooth hypersurface of degree $\delta \geq 1$. Then, we have $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < d$.*

Proof. Let $\mathbb{P} = \mathbb{P}^{d+1}$. We have a short exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}}(-\delta) \to \mathcal{O}_{\mathbb{P}} \to \mathcal{O}_X \to 0.$$

We have $H^i(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m)) = 0$ for all $0 < i < d+1$ and all $m \in \mathbb{Z}$. Now, the long exact sequence shows $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < d$. □

Fact 2.5. *Let $X \to \operatorname{Spec} \mathbb{C}$ and $Y \to \operatorname{Spec} \mathbb{C}$ be smooth, proper, and irreducible. Then, $X \times Y \to \operatorname{Spec} \mathbb{C}$ is smooth, proper, and irreducible, and we have*

$$\Omega_{X\times Y}^{d_1+d_2} \cong p_X^* \Omega_X^{d_1} \otimes p_Y^* \Omega_Y^{d_2}$$

for the two projections $p_X \colon X \times Y \to X$ and $p_Y \colon X \times Y \to Y$. In particular, the product of two algebraic Calabi–Yau manifolds is an algebraic Calabi–Yau manifold.

Example 2.6. (1) Let $E = \{f = y^2z - x^3 - x^2z - z^3 = 0\} \subset \mathbb{P}^2$. This is a smooth *elliptic curve*. Since $\deg(f) = 3$, we have $\Omega_E^1 \cong \mathcal{O}_E$ so that E is an algebraic Calabi–Yau manifold. The topological space E^{top} is a real torus, i.e., $E^{\mathrm{top}} \approx S^1 \times S^1$.

(2) The product $E \times E$ is an algebraic Calabi–Yau manifold. It is an *Abelian surface* and not strict. Namely, we have $\dim H^1(E \times E, \mathcal{O}_{E\times E}) = 2$.

(3) The scheme $C = \{f = x^4 + y^4 + 10xyz^2 - z^4 = 0\} \subset \mathbb{P}^2$ is a smooth quartic curve. We have $\Omega_C^1 \cong \mathcal{O}_C(1)$ so that C is not Calabi–Yau. The topological space C^{top} is a Riemann surface of genus 2, i.e., it has two holes.

(4) Let $Q = \{f = x^4 + y^4 + z^4 + w^4 = 0\} \subset \mathbb{P}^3$. This is a smooth quartic surface. We have $\delta = \deg(f) = 4$ and hence $\Omega_Q^2 \cong \mathcal{O}_Q(-2-2+4) = \mathcal{O}_Q$ so that it is an algebraic Calabi–Yau manifold. We have $H^1(Q, \mathcal{O}_Q) = 0$ so that Q is strict.

(5) Let $F = \{f = x^5 + y^5 + z^5 + w^5 + u^5 = 0\} \subset \mathbb{P}^4$. It is a smooth hypersurface; we have $\Omega_F^3 \cong \mathcal{O}_F(-3-2+5) = \mathcal{O}_F$ so that it is an algebraic Calabi–Yau manifold. It is usually called the *Fermat quintic threefold*. We have $H^1(F, \mathcal{O}_F) = 0$ and $H^2(F, \mathcal{O}_F) = 0$ so that F is strict.

To obtain some information about the topology of F^{top}, one computes the Hodge numbers. The result is

$$h^1(F, \Omega_F^1) \coloneqq \dim H^1(F, \Omega_F^1) = 1 \quad \text{and} \quad h^2(F, \Omega_F^1) = 101.$$

Hodge symmetry and the Hodge decomposition then allows us to compute the *Betti numbers* as

$$b_2(F) \coloneqq \dim H^2(F^{\mathrm{top}}, \mathbb{C}) = h^0(F, \Omega_F^2) + h^1(F, \Omega_F^1) + h^2(F, \mathcal{O}_F) = 1$$

and

$$\begin{aligned} b_3(F) &= \dim H^3(F^{\mathrm{top}}, \mathbb{C}) \\ &= h^0(F, \Omega_F^3) + h^1(F, \Omega_F^2) + h^2(F, \Omega_F^1) + h^3(F, \mathcal{O}_F) \\ &= 2 + 2 \cdot h^2(F, \Omega_F^1) = 204. \end{aligned}$$

They are invariants of F^{top}. ◊

Lemma 2.7. *Let $X, Y \to \operatorname{Spec} \mathbb{C}$ be two algebraic Calabi–Yau manifolds of positive dimension. Then, $X \times Y$ is not strict.*

Proof. We have the *Künneth formula*

$$H^k(X \times Y, \mathcal{O}_{X\times Y}) \cong \bigoplus_{i+j=k} H^i(X, \mathcal{O}_X) \otimes H^j(Y, \mathcal{O}_Y).$$

Then, for $d_X = \dim(X)$, we find that $H^{d_X}(X, \mathcal{O}_X) \otimes H^0(Y, \mathcal{O}_Y) \neq 0$ is a direct summand of $H^{d_X}(X \times Y, \mathcal{O}_{X\times Y})$. □

The hypersurface construction yields one family of strict algebraic Calabi–Yau manifolds in every dimension. If $X, Y \subset \mathbb{P}^n$ are smooth hypersurfaces of the same degree, one may prove that $X^{\mathrm{top}} \approx Y^{\mathrm{top}}$. The following question will guide our introduction.

Question 2.8. *Fix $d \geq 1$. How many strict algebraic Calabi–Yau manifolds of dimension d are there such that the underlying Euclidean topological spaces are pairwise not homeomorphic? Is the number finite or infinite?*

For $d = 1$, all strict algebraic Calabi–Yau manifolds are elliptic curves and hence real tori. Similarly, for $d = 2$, all strict algebraic Calabi–Yau manifolds are K3 surfaces and hence pairwise homeomorphic. In the rest of the talk, we will discuss $d = 3$.

3 Complete intersections

We construct additional strict algebraic Calabi–Yau manifolds. Let $f_1, \ldots, f_r \in \mathbb{C}[x_0, \ldots, x_n]$ be homogeneous polynomials. We say that

$$X = \{f_1 = \ldots = f_r = 0\} \subset \mathbb{P}^n$$

is a *complete intersection* if $\dim(X) = n - r$. In this case, every irreducible component of X has dimension $d \coloneqq n - r$.

Fact 3.1. *Let $X \subset \mathbb{P}^n$ be a smooth complete intersection of dimension $d = n - r$. Then, we have $\Omega_X^d \cong \mathcal{O}_X(-n - 1 + \delta)$ for $\delta = \sum_{j=1}^r \deg(f_j)$.*

Fact 3.2. *Let $X \subset \mathbb{P}^n$ be a complete intersection of dimension $d = n - r \geq 1$. Then, X is connected, and we have $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < d$. If X is smooth, then it is irreducible.*

Example 3.3. (1) Let $f_1 = xyz - (u^3 + v^3 + w^3)$ and $f_2 = uvw - (x^3 + y^3 + z^3)$. Then, $X = X(3,3) = \{f_1 = f_2 = 0\} \subset \mathbb{P}^5$ is a smooth complete intersection of dimension $d = 5 - 2 = 3$. We have $\Omega^3_X \cong \mathcal{O}_X(-5 - 1 + 3 + 3) = \mathcal{O}_X$ so that it is an algebraic Calabi–Yau manifold. It is strict, and we have $h^1(X, \Omega^1_X) = 1$ and $h^2(X, \Omega^1_X) = 73$. Thus, we have $b_2(X) = 1$ and $b_3(X) = 2 + 2 \cdot 73 = 148$ for the Betti numbers.

(2) Let $f_1 = (x^4 + y^4 + z^4 + w^4 + u^4 + v^4) + xyzw$ and $f_2 = u^2 + v^2 + xy + zw$. Then, $X = X(4,2) = \{f_1 = f_2 = 0\} \subset \mathbb{P}^5$ is a smooth complete intersection of dimension $d = 5 - 2 = 3$. We have $\Omega^3_X \cong \mathcal{O}_X(-5 - 1 + 4 + 2) = \mathcal{O}_X$ so that it is an algebraic Calabi–Yau manifold. It is strict, and we have $h^1(X, \Omega^1_X) = 1$ and $h^2(X, \Omega^1_X) = 89$. In particular, $b_2(X) = 1$ and $b_3(X) = 2 + 2 \cdot 89 = 180$.

(3) It is possible to find homogeneous polynomials $f_1, f_2, f_3 \in \mathbb{C}[x_0, \ldots, x_6]$ with $\deg(f_1) = 3$ and $\deg(f_2) = \deg(f_3) = 2$ such that $X = X(3,2,2) = \{f_1 = f_2 = f_3 = 0\} \subset \mathbb{P}^6$ is a smooth complete intersection of dimension $d = 6 - 3 = 3$. We have $\Omega^3_X \cong \mathcal{O}_X(-6 - 1 + 2 + 2 + 3) = \mathcal{O}_X$ so that it is an algebraic Calabi–Yau manifold. It is strict, and we have $h^1(X, \Omega^1_X) = 1$ and $h^2(X, \Omega^1_X) = 73$, the same values as for $X(3,3)$.

One may show that $X(3,3)^{\mathrm{top}} \not\approx X(3,2,2)^{\mathrm{top}}$ by considering the integral cohomology rings $H^\bullet(X(3,3)^{\mathrm{top}}, \mathbb{Z})$ and $H^\bullet(X(3,2,2)^{\mathrm{top}}, \mathbb{Z})$, which are not isomorphic as commutative rings. Namely, for any smooth complete intersection $i\colon X \subset \mathbb{P}^n$, we have $\int_X i^*[H]^d = \int_{\mathbb{P}^n} [H]^d \cdot [X]$ for the hyperplane class $[H] \in H^2(\mathbb{P}^n, \mathbb{Z})$, and moreover, $[X] = \prod_{j=1}^r \deg(f_j) \cdot [H]$. Thus, $\int_X i^*[H]^d = \prod_{j=1}^r \deg(f_j)$.

(4) It is possible to find four homogeneous polynomials of degree 2 in $\mathbb{C}[x_0, \ldots, x_7]$ such that $X = X(2,2,2,2) = \{f_1 = f_2 = f_3 = f_4 = 0\} \subset \mathbb{P}^7$ is a smooth complete intersection of dimension $d = 7-4 = 3$. We have $\Omega^1_X \cong \mathcal{O}_X(-7-1+2+2+2+2) = \mathcal{O}_X$ so that it is an algebraic Calabi–Yau manifold. It is strict, and we have $h^1(X, \Omega^1_X) = 1$ and $h^2(X, \Omega^1_X) = 65$ so that $b_2(X) = 1$ and $b_3(X) = 2 + 2 \cdot 65 = 132$. ♢

Lemma 3.4. *These are all algebraic Calabi–Yau manifolds that one can obtain as complete intersections in $\mathbb{P}^n$.*

Proof. To be interesting, we must have $\deg(f_j) \geq 2$, so we have $\delta \geq 2r$. On the other hand, the CY condition yields $\delta = n + 1$, so we have $d + r + 1 \geq 2r$ and hence $r \leq d + 1$. □

One may replace the ambient space $\mathbb{P}^n$ with other varieties:

(1) $\mathbb{P} = \mathbb{P}^{n_1} \times \mathbb{P}^{n_2} \times \ldots \times \mathbb{P}^{n_\ell}$;

(2) $\mathbb{P} = \mathbb{P}(a_0, \ldots, a_n)$ a weighted projective space;

(3) $\mathbb{P} = \mathbb{P}(\Delta)$ a Gorenstein toric Fano variety.

Furthermore, if $X \subset \mathbb{P}$ is not smooth, it is often possible to resolve its singularities without destroying the Calabi–Yau property. With these strategies, one can find more than 30,000 strict algebraic Calabi–Yau threefolds X such that their X^{top} are pairwise not homeomorphic.

4 Degenerating families

It appears that constructing further algebraic Calabi–Yau manifolds as complete intersections in some ambient space is unlikely to decide the question whether there are finitely

many or infinitely many pairwise non-homeomorphic algebraic Calabi–Yau manifolds of a given dimension d. We now introduce another technique that potentially can be used to construct infinitely many pairwise non-homeomorphic Calabi–Yau manifolds.

Example 4.1. Let us consider

$$E = \{y^2z - x^3 - x^2z - tz^3 = 0\} \subset \mathbb{P}^2 \times \mathbb{A}^1_t,$$

together with the morphism $\phi\colon E \to \mathbb{A}^1_t$ induced by the second projection. This morphism is flat and proper. For a general $t \in \mathbb{A}^1_t$, the fiber $E_t = \phi^{-1}(t) \subset \mathbb{P}^2$ is a smooth elliptic curve. In particular, $E_t^{\mathrm{top}} \approx S^1 \times S^1$. For $t = 0$, the fiber $E_0 = \phi^{-1}(0)$ is a *nodal cubic*. Topologically, E_0^{top} is a *pinched torus*. Thus, E_0 is not an algebraic Calabi–Yau manifold because it is not smooth, but the family $\phi\colon E \to \mathbb{A}^1_t$ shows that E_0 is somehow close to being an algebraic Calabi–Yau manifold. ♢

The nodal cubic E_0 is not only a pinched torus. It is also obtained by gluing two points in $\mathbb{P}^1$ to each other. This suggests the following strategy to find more algebraic Calabi–Yau manifolds:

(1) construct a scheme V by such means as gluing $\mathbb{P}^1$ to itself in two points;

(2) show that there is a family $\phi\colon X \to \mathbb{A}^1_t$ with $V \cong X_0 = \phi^{-1}(0)$ and such that $X_t = \phi^{-1}(t)$ is an algebraic Calabi–Yau manifold for a general $t \in \mathbb{A}^1_t$.

Motivated by Question 2.8 and by the strategy, Hashimoto and Sano have constructed an infinite sequence of singular schemes $X_0(a)$:

Construction 4.2 (Hashimoto–Sano 2023)**.** Let $a \geq 1$ be a positive integer.

(a) Let $Y(a) = \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$, and let $S(a) \subset Y(a)$ be a very general hypersurface of type $(2, 2, 2)$.

(b) We have another irreducible smooth projective threefold $W(a)$ and a closed immersion $S(a) \to W(a)$. The scheme $W(a)$ depends on a.

(c) We set $X_0(a) \coloneqq Y(a) \sqcup_{S(a)} W(a)$, the pushout in the category of schemes.

(d) $X_0(a)$ has two irreducible components $Y(a)$ and $W(a)$; their intersection is $Y(a) \cap W(a) = S(a)$. Any point $p \in S(a) \subset X_0(a)$ has an étale neighborhood of the form $\operatorname{Spec} \mathbb{C}[x, y]/(xy) \times \mathbb{A}^2$.

(e) $X_0(a) \to \operatorname{Spec} \mathbb{C}$ is proper but not projective. ♦

5 Formal deformations

Let $V = X_0(a)$ be one of the singular schemes constructed by Hashimoto and Sano. We would like to show the existence of a flat and proper morphism of schemes $\phi\colon X \to \mathbb{A}^1_t$ whose special fiber is $\phi^{-1}(0) \cong X_0(a)$, and such that X_t is an algebraic Calabi–Yau manifold for a general $t \in \mathbb{A}^1_t$.

Assume that such a family $\phi\colon X \to \mathbb{A}^1_t$ is given. For $k \geq 0$, let $A_k = \mathbb{C}[t]/(t^{k+1})$, and let $S_k = \operatorname{Spec} A_k$. Topologically, each S_k is a point, but the ring of functions depends on

k. For $k \leq \ell$, we have a closed immersion $S_k \to S_\ell$. Moreover, we have a closed immersion $S_k \to \mathbb{A}^1_t$, and $S_0 \to \mathbb{A}^1_t$ is the embedding of $0 \in \mathbb{A}^1_t$. Then, we can form the diagram

$$\begin{array}{ccccccc} X_0 & \longrightarrow & X_1 & \longrightarrow & X_2 & \longrightarrow \quad \dots & \longrightarrow X \\ \downarrow{\scriptstyle \phi_0} & & \downarrow{\scriptstyle \phi_1} & & \downarrow{\scriptstyle \phi_2} & & \downarrow{\scriptstyle \phi} \\ S_0 & \longrightarrow & S_1 & \longrightarrow & S_2 & \longrightarrow \quad \dots & \longrightarrow \mathbb{A}^1_t \end{array}$$

with all squares Cartesian. We say that $(\phi_k \colon X_k \to S_k)_{k \geq 0}$ is a *formal flat deformation* of $V = X_0$. Conversely, it is generally easier to construct formal flat deformations than to construct a family $\phi \colon X \to \mathbb{A}^1_t$. Then, one can use the following *analytic* existence result:

Fact 5.1. *Let $V = X_0 \to \operatorname{Spec} \mathbb{C}$ be proper, and let $(\phi_k \colon X_k \to S_k)_{k \geq 0}$ be a formal flat deformation. For every $c \geq 0$, there is a flat and proper morphism of complex analytic spaces $\phi \colon \mathcal{X} \to \Delta$ to a small disk $\Delta = \{z \in \mathbb{C} \mid |z| < \varepsilon\}$ such that we have a Cartesian square*

$$\begin{array}{ccc} X_c^{\mathrm{an}} & \longrightarrow & \mathcal{X} \\ \downarrow{\scriptstyle \phi_k} & & \downarrow{\scriptstyle \phi} \\ S_c^{\mathrm{an}} & \longrightarrow & \Delta \end{array}$$

of complex analytic spaces.

Note that it is not necessarily possible to find one $\phi \colon \mathcal{X} \to \Delta$ which works for all $c \geq 0$.

There is no analog of this fact that yields a family over $\mathbb{A}^1_t$ or more generally over a smooth affine algebraic curve. For now, we will be satisfied with the analytic existence result. For instance, it could be the case that $\mathcal{X}_t$ is projective, and then it is algebraic even though it is not the fiber of a family of schemes.

Let $\phi \colon \mathcal{X} \to \Delta$ be a family obtained from Fact 5.1. Then, it is not necessarily true that $\mathcal{X}_t$ is smooth for $t \neq 0$, even for very small $0 < |t| \ll \varepsilon$.

Definition 5.2. A formal flat deformation $(\phi_k \colon X_k \to S_k)_{k \geq 0}$ is a *formal smoothing* if there is some $m \geq 0$ such that $t^m \in \operatorname{Fitt}_d(\Omega^1_{X_m / S_m})$ for $d = \dim(X_0)$ and the d-th Fitting ideal of $\Omega^1_{X_m / S_m}$. ♦

At this point, it is not necessary to understand what precisely this definition means. The important point is the following: If $(\phi_k \colon X_k \to S_k)_{k \geq 0}$ is a formal smoothing, m is the value from the definition, and $\phi \colon \mathcal{X} \to \Delta$ is the family from Fact 5.1 for $c \geq m$, then $\mathcal{X}_t$ is smooth for $0 < |t| \ll \varepsilon$.

6 Unobstructedness

Let $V = X_0(a)$ be one of the schemes constructed by Hashimoto and Sano. The most important question now is if V admits a formal smoothing $(\phi_k \colon X_k \to S_k)_{k \geq 0}$. First, we review a peculiar property of algebraic Calabi–Yau manifolds which suggests that this might be possible.

Theorem 6.1 (Special case of Bogomolov–Tian–Todorov)**.** *Let $\phi_0 \colon X_0 \to \operatorname{Spec} \mathbb{C}$ be an algebraic Calabi–Yau manifold. Fix $k \geq 0$, and assume that we have a Cartesian square*

$$\begin{array}{ccc} X_0 & \longrightarrow & X_k \\ \downarrow{\scriptstyle \phi_0} & & \downarrow{\scriptstyle \phi_k} \\ S_0 & \longrightarrow & S_k \end{array}$$

where $\phi_k \colon X_k \to S_k$ is flat. Then, we can extend the diagram to

$$\begin{array}{ccccc} X_0 & \longrightarrow & X_k & \longrightarrow & X_{k+1} \\ \downarrow{\scriptstyle \phi_0} & & \downarrow{\scriptstyle \phi_k} & & \downarrow{\scriptstyle \phi_{k+1}} \\ S_0 & \longrightarrow & S_k & \longrightarrow & S_{k+1} \end{array}$$

where the right-hand square is Cartesian, and where $\phi_{k+1} \colon X_{k+1} \to S_{k+1}$ is flat.

Thus, every flat deformation $\phi_m \colon X_m \to S_m$ of an algebraic Calabi–Yau manifold can be extended to a formal flat deformation.

The theorem provides a blueprint for constructing a formal smoothing of $V = X_0(a)$. There are however two serious issues:

(1) Suppose that the theorem is true for $V = X_0(a)$. Then, we obtain a formal deformation $(\phi_k \colon X_k \to S_k)_{k \geq 0}$. However, the formal deformation might not be a formal smoothing. A trivial example is given by $X_k = X_0 \times S_k$.

(2) The theorem is indeed true for some mildly singular schemes once an appropriate analog of the Calabi–Yau condition $\Omega_X^d \cong \mathcal{O}_X$ is imposed. I don't know if it is true for $V = X_0(a)$, but it is false for some very similar schemes. Thus, we should assume for now that the theorem is false for $V = X_0(a)$.

Both problems are solved by logarithmic algebraic geometry. For today, we simply note that there is a category of *logarithmic schemes* $\mathrm{Sch}^{\log}$, and that it admits a forgetful functor $\mathrm{Sch}^{\log} \to \mathrm{Sch}$. Each S_k has a canonical enhancement to a log scheme $S_k^\dagger$. The log scheme $S_0^\dagger$ is called the *standard log point*.

Fact 6.2. *Let $V = X_0 = X_0(a)$. Then:*

(1) *we have an enhancement $\phi_0 \colon X_0^\dagger \to S_0^\dagger$ to a morphism of log schemes;*

(2) *this morphism is logarithmically smooth;*

(3) *the sheaf of log differential forms $\Omega^{1,\log}_{X_0/S_0}$ is a vector bundle of rank 3;*

(4) $\Omega^{3,\log}_{X_0/S_0} \cong \mathcal{O}_{X_0}$.

Now, Theorem 6.1 is true for log smooth deformations of $\phi_0 \colon X_0(a)^\dagger \to S_0^\dagger$ and the closed embedding $S_k^\dagger \to S_{k+1}^\dagger$. Thus, we obtain a formal log smooth deformation $(\phi_k \colon X_k(a)^\dagger \to S_k^\dagger)_{k \geq 0}$. Moreover, the underlying formal flat deformation is a formal smoothing—this holds for any formal log smooth deformation, unlike the case of flat deformations, where singularities can be preserved. We obtain a flat and proper holomorphic family $\phi \colon \mathcal{X} \to \Delta$. The geometry of this family has been studied by Hashimoto and Sano:

Theorem 6.3 (Hashimoto–Sano 2023)**.** *Let $X_0(a)$ be as above. Then, there is a flat and proper holomorphic map $\phi\colon \mathcal{X} \to \Delta$ from a complex manifold $\mathcal{X}$ to a small disk*

$$\Delta = \{z \in \mathbb{C} \mid |z| < \varepsilon\}$$

with the following properties:

(a) $\phi^{-1}(0) \cong X_0(a)$*;*

(b) *for $t \neq 0$, the fiber $\mathcal{X}_t = \phi^{-1}(t)$ is a simply connected strict compact complex Calabi–Yau manifold of dimension* 3*;*

(c) *for $t \neq 0$, we have $b_2(\mathcal{X}_t) \coloneqq \dim\, H^2(\mathcal{X}_t, \mathbb{C}) = a + 3$;*

(d) *for $t \neq 0$, the fiber $\mathcal{X}_t$ is neither algebraic nor does it admit a Kähler metric.*

Thus, we obtain infinitely many pairwise non-homeomorphic complex Calabi–Yau threefolds, but they are not algebraic and do not even have a Kähler metric. For algebraic Calabi–Yau manifolds, the question is still open.